

Problem set 1: Basics

A.Demand

1. Please determine for each of the following examples, whether the products are substitutes, complements, or neither and explain in 1-2 sentences why.
 - a. Coffee and Tea
 - b. Coffee and Sugar
 - c. Coffee and Cigarettes
 - d. Paper and Computer
 - e. Houses for sale and apartments for rent.
 - f. Butter and butterflies
 - g. Gold and Bitcoin
2. Please circle all the correct answers (there might be more than one) for the following multiple-choice questions.
 - a. *Willingness to pay is the ____ amount that ____ are willing to pay in order to ____ a given product.*
 - i. Lowest, Consumers, Buy.
 - ii. Highest, Producers, Sell
 - iii. Lowest, Producers, Buy
 - iv. Highest, Consumers, Buy
 - b. *Suppose product A and product B are ____ . This means that when the price of A ____ then the demand for B ____.*
 - i. Substitutes, increases, increases.
 - ii. Complements, increases, increases.
 - iii. Substitutes, decreases, increases.
 - iv. Complements, decrease, increases.
3. Explain in 1-2 sentences what is the “law of demand”
4. Suppose that the following demanded quantities are given. Draw the demand function.

Price	Quantity Demanded
10	50
20	40
30	30
40	20
50	10

5. Suppose that the following quantities are given. Draw the demand function and determine (without calculation) the elasticity of demand.

Price	Quantity Demanded
10	10
10	20
10	30
10	40
10	50

6. Suppose that the following quantities are given. Draw the demand function and determine (without calculation) the elasticity of demand.

Price	Quantity Demanded
10	20
20	20
30	20
40	20
50	20

7. Suppose that a hand sanitizer costs 50\$ and that for this price, each consumer is willing to buy 5 units. One day, the price increases to 70\$, and consumers are now willing to buy only 3 units each.
- Calculate the elasticity of demand at the point of the original price and quantity.
 - Is the demand elastic, inelastic, or unit-elastic?
 - Suppose that after the price increase, there is a price decrease: each sanitizer is now sold for 40\$ and consumers are then willing to buy 4 units. Calculate the elasticity at the point of the new(est) price and quantity (i.e. after the decrease). Is the demand elastic, inelastic, or unit-elastic? (note: use the numbers given for the “old” price of 50\$; i.e. ignore the “new” price of 70\$ here).
8. Suppose that a firm has the following inverse demand function:¹ $p = 10 - 2q$. Calculate the point elasticity of demand (PED) when the price equals 20.

¹ An inverse demand function is expressed by the price, p , on the LHS and the quantity demanded, q , on the RHS.

B. Supply

1. Explain the difference between a fixed cost and a variable cost
2. Explain the difference between an average cost and a marginal cost
3. What is the production-possibility frontier (“PPF”)?
4. Determine whether the following statements are correct or incorrect and explain why in 1-2 sentences:
 - a. If the firm produces at a point which is on the production-possibility frontier, then there is full employment.
 - b. if the firm produces below the PPF, there is unemployment.
 - c. If the price is higher than the firm’s marginal cost, then the firm should not produce.
 - d. If the price is lower than the firm’s average cost, the firm should produce in the short run but possibly not in the long-run.
5. Suppose you have 100 workers: 50 workers of type A and 50 of type B. You can produce two products: Flowers and Chairs. To produce each product, 25 workers are needed. The output table is:

	Flowers	Chairs
Worker A	1	5
Worker B	2	8

Which type of worker should be allocated for producing which product?

6. Suppose that the supply curve’s formula is $p = 12 + 3q$.
 - a. Explain why there is a ‘plus’ sign (rather than a ‘minus’ sign) on the RHS.
 - b. If the price in the market is 42, how many units are firms willing to produce?

C.Equilibrium

1. The demand and supply functions for Coffee are given by:

$$Q^d = 1000 - 5p$$
$$Q^s = 200 + 3p$$

Where Q^d is the quantity demanded and Q^s is the quantity supplied.

- a. Plot the supply and demand curves on a graph and show where the equilibrium occurs.
 - b. Calculate the equilibrium price and quantity (using algebra).
2. A manufacturer in a perfectly competitive market produces product X using one input factor:
I. Suppose that price of I increases but nothing else changes. The manufacturer will then:
- (1) Increase the produced quantity
 - (2) Raise the price of product X
 - (3) Decrease the produced quantity
 - (4) Answers (1) and (2) are both correct
 - (5) Answers (1) and (3) are both correct
3. Here are two claims about a firm operating in a perfectly competitive market:
- (1) If the variable cost for producing 3 units is 300\$, the average cost per unit for producing 4 units is 100\$ and the firm has a fixed cost of 80\$, then the marginal cost is 20\$.
 - (2) The firm has an increasing marginal cost function and produces 20 units. If the firm increase the price in which it sell a unit by 3\$, the profits of the firm will increase by 60\$.
- Are these claims correct? If not, explain why not.
4. A firm is a monopoly. Therefore:
- (1) It has a downward-sloping marginal revenue curve.
 - (2) It has an upward-sloping marginal revenue curve
 - (3) It has a fixed marginal revenue.
5. A firm is a monopoly. Therefore its marginal revenue curve is:
- (1) Below the demand curve.
 - (2) Above the demand curve.
 - (3) Identical to the demand curve.

6. Suppose that the price of coffee beans increases and the demand for Capuccino increases. What will happen to the price of a cup of Capuccino?
 - (1) The price will increase
 - (2) The price will decrease
 - (3) The price will not change
 - (4) There is not enough information to determine whether the price will increase or decrease.
7. Suppose that the price of tea increases and the supply of coffee increases. What will happen to the price of a cup of Capuccino?
 - (1) The price will increase
 - (2) The price will decrease
 - (3) The price will not change
 - (4) There is not enough information to determine whether the price will increase or decrease.
8. Suppose that the price of a keyboard increases and the supply of computers increases. What will happen to the price of a computer?
 - (1) The price will increase
 - (2) The price will decrease
 - (3) The price will not change
 - (4) We cannot determine whether the price will increase or decrease.
9. Suppose that the demand for antibiotics is completely inelastic. A new and cheaper way to produce antibiotics is discovered. What will happen to the price of antibiotics? What will happen to the quantity sold?
 - (1) The price will increase and the quantity will decrease.
 - (2) The price will increase and the quantity will increase.
 - (3) The price will not change but the quantity will increase.
 - (4) The price will decrease but the quantity will not change.
10. Suppose that the demand for cola is completely elastic. The wage of workers decreases. What will happen to the price of cola? What will happen to the quantity sold?
 - (1) The price will increase and the quantity will decrease.
 - (2) The price will increase and the quantity will increase.
 - (3) The price will not change but the quantity will increase.
 - (4) The price will decrease but the quantity will not change.

D.Efficiency - basics

1. Explain the difference between Pareto efficiency and Kaldor-Hicks efficiency.
2. Which of the following distributions of money between “A” and “B” is Pareto *inefficient*? Which distribution is Kaldor-Hicks *efficient*?

Distribution 1: A has 20. B has 20.

Distribution 2: A has 10. B has 20.

Distribution 3: A has 15. B has 25

Distribution 4: A has 100. B has 5.

3. I have A car, but I like scooters more. You have a scooter, but you like cars more. If we exchange my car for your scooter, this is...
 - (1) Inefficient
 - (2) Pareto efficient
 - (3) Kaldor-Hicks efficient
 - (4) Not enough information to determine
4. We have the choice between 4 distributions:
 - (1) I get 5\$, you get 5\$
 - (2) I get 3\$, you get 10\$.
 - (3) I get 15\$, you get 2\$.
 - (4) I get 10\$, you get 7\$

Which distribution is pareto inefficient? Which distribution is Kaldor-hicks efficient?

5. We have the choice between 4 distributions:
 - (1) I get 1\$, you get 1\$
 - (2) I get 2\$, you get 7\$.
 - (3) I get 9\$, you get 2\$.
 - (4) I get 10\$, you get 10\$

Which distributions are pareto efficient? Which distribution is Kaldor-hicks efficient?

6. We have the choice between 3 distributions:
 - (1) A gets 3\$, B gets 8\$.
 - (2) A gets 100\$, B gets 2\$,
 - (3) A gets 4\$ B gets 4\$

Which distribution is optimal according to Rawls?

7. A firm's CEO promises not to embezzle money, but the board of directors fears he might do so anyway. The firm is trying to decide whether to create an algorithm that automatically discloses embezzlements to the police. Discuss what is the tradeoff that the firm faces from an ex-ante and an ex-post efficiency perspective.

E.Surplus

1. Draw a set of standard supply and demand curves and mark the consumer surplus and the supplier surplus.
2. What is the consumer surplus when the demand is (1) perfectly elastic and (2) perfectly inelastic?
3. Explain why the supplier surplus increases as the supply becomes inelastic.